

House Price Prediction using Model Comparison

Task 2

Introduction

This project extends the basic House Price Prediction system developed in Task 1 by incorporating feature scaling, multiple regression models, and performance comparison techniques. The goal was to identify the best-performing model for predicting house prices using the California Housing Dataset.

Dataset Description

The California Housing Dataset contains 20,640 records and 8 input features including Median Income, House Age, Average Rooms, Population, Latitude, and Longitude. The target variable is House Price.

Data Preprocessing

Feature scaling was performed using StandardScaler to ensure that all features were on a similar scale. This helps machine learning models learn more effectively and improves stability during training.

Models Used

Three regression algorithms were trained and evaluated:

- Linear Regression
- Ridge Regression
- Decision Tree Regressor

Model Evaluation

The models were compared using:

- RMSE (Root Mean Squared Error)
- R^2 Score

A comparison table was generated to identify the best-performing model.

	RMSE	R2 Score
Linear Regression	0.745581	0.575788
Ridge Regression	0.745554	0.575819
Decision Tree	0.724234	0.599732

Figure 1: Model Comparison Results

Visualization

An Actual vs Predicted House Prices graph was created using the best-performing model.

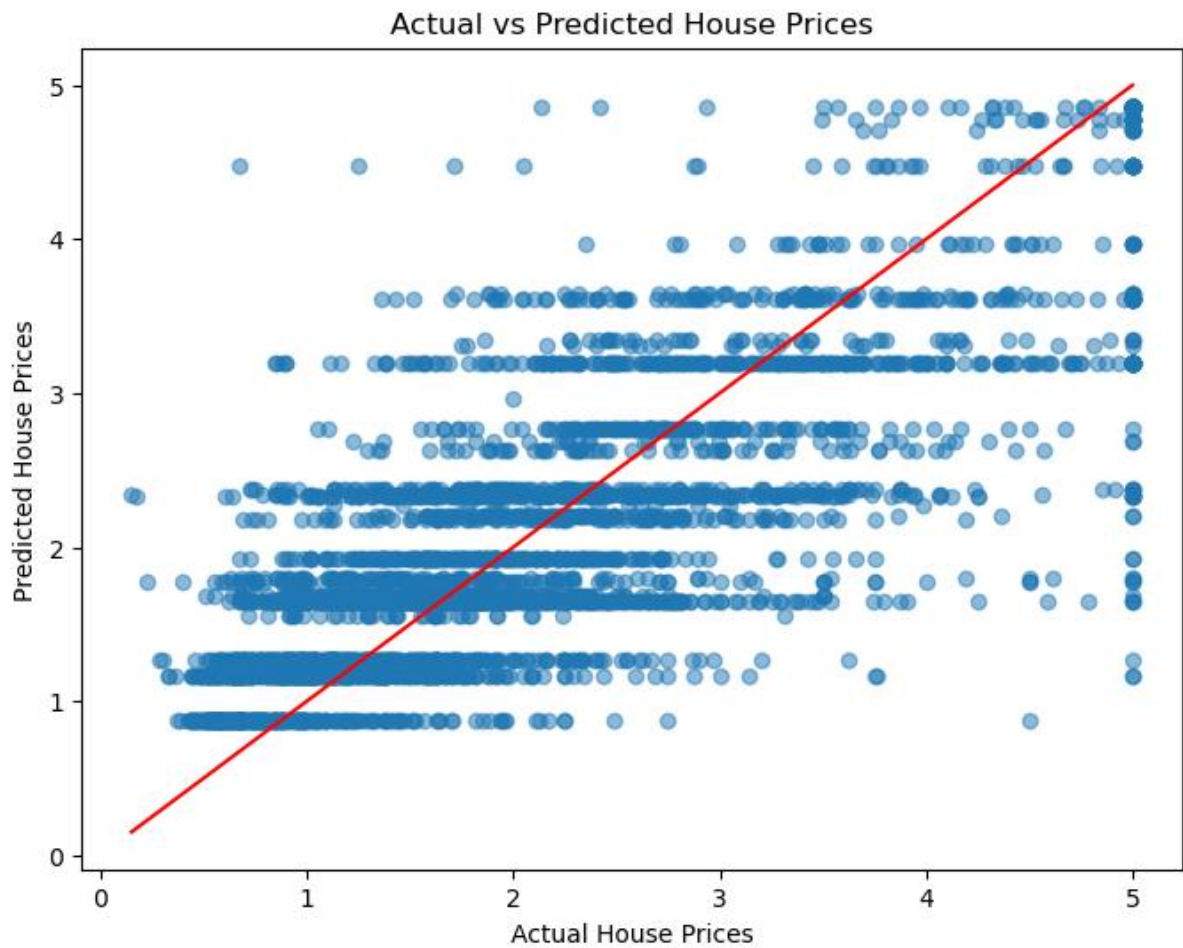


Figure 2: Actual vs Predicted House Prices

Conclusion

Conclusion

In this project, multiple regression models were trained and compared using the California Housing Dataset. Feature scaling was applied to improve data consistency and model performance.

Three models were evaluated: Linear Regression, Ridge Regression, and Decision Tree Regressor. The models were compared using RMSE and R^2 Score metrics.

Among all models, the Decision Tree Regressor achieved the best performance with an RMSE of 0.724 and an R^2 Score of 0.600. Therefore, it was selected as the final model for house price prediction.

This project provided practical experience in feature scaling, model comparison, performance evaluation, and model selection, which are essential steps in a professional Machine Learning workflow.

Future Improvements

- Hyperparameter Tuning
- Random Forest Regression
- Gradient Boosting Models
- Cross Validation
- Feature Engineering